

Experiment No:4

Demonstrate use of looping Statement:a)'While' loop b)'for' loop c)Nested loop

In []: *# For Loops*

```
In [7]: fruits = ["apple", "banana", "cherry"]
for x in fruits:
    print(x)
```

apple
banana
cherry

```
In [2]: for x in "banana":
        print(x)
```

b
a
n
a
n
a

In []: *# Break Statement*

```
In [5]: fruits = ["apple", "banana", "cherry"]
for x in fruits:
    if x == "banana":
        break
    print(x)
    break
```

apple
banana

```
In [8]: fruits = ["apple", "banana", "cherry"]
        for x in fruits:
            if x == "banana":
                break
            print(x)
```

apple

```
In [9]: # Continue Statement
```

```
In [10]: fruits = ["apple", "banana", "cherry"]
         for x in fruits:
             if x == "banana":
                 continue
             print(x)
```

apple
cherry

```
In [11]: # Range Function
```

```
In [12]: for x in range(6):
         print(x)
```

0
1
2
3
4
5

```
In [14]: for x in range(2,30,3):  
         print(x)
```

```
2  
5  
8  
11  
14  
17  
20  
23  
26  
29
```

```
In [17]: # Exercise
```

```
In [16]: for x in range(1,5):  
         for y in range(x):  
             print("*",end="")  
         print()
```

```
*  
**  
***  
****
```

```
In [19]: for x in range(1,5):  
         for y in range(x, 0, -1):  
             print(y, end="")  
         print()
```

```
1  
21  
321  
4321
```

```
In [20]: # Else in for Loop
```

```
In [22]: for x in range(6):  
        print(x)  
        else:  
            print("Finaily finished!")
```

```
0  
1  
2  
3  
4  
5  
Finaily finished!
```

```
In [24]: for x in range(6):  
        if x ==3:  
            break  
        print(x)  
        else:  
            print("Finaliy Finished!")
```

```
0  
1  
2
```

```
In [ ]: # Nested Loops
```

```
In [28]: adj = ["red", "big", "tasty"]
         fruits = ["apple", "banana", "cherry"]
```

```
         for x in adj:
             for y in fruits:
                 print(x, y)
```

```
red apple
red banana
red cherry
big apple
big banana
big cherry
tasty apple
tasty banana
tasty cherry
```

```
In [29]: # The pass statement
```

```
In [30]: for x in [0, 1, 2]:
         pass
```

```
In [31]: # While Loops
```

```
In [32]: i = 1
         while i<6:
             print(i)
             i +=1
```

```
1
2
3
4
5
```

```
In [ ]: # Break Statement
```

```
In [33]: i = 1
        while i<6:
            print(i)
            if i == 3:
                break
            i +=1
```

1
2
3

```
In [34]: # Continue Statement
```

```
In [1]: i = 0
        while i < 6:
            i += 1
            if i == 3:
                continue
            print(i)
```

1
2
4
5
6

```
In [2]: # else statement
```

```
In [3]: i = 1
while i < 6:
    print(i)
    i += 1
else:
    print("i is no longer less than 6")
```

1

2

3

4

5

i is no longer less than 6